Member	Role	Key Deliverables
Member 1 – Data Architect	Leads **database design** and normalization (ERD, table structure, PK/FK setup).	• Create ER diagram (Visio, Lucidchart, or Draw.io).• Define entity attributes & relationships.• Ensure 3NF normalization.
Member 2 – SQL Developer	Handles **implementation** and **queries** in SQL Server.	• Write CREATE TABLE scripts.• Populate with sample data.• Prepare 10 SQL queries (4 with joins, 4 with aggregation/subquery).
Member 3 – Presenter / Analyst	Handles **slides** and **business insight narrative**.	• Prepare slides for objectives, use case, assumptions, and "managerial insights."• Coordinate timing and flow of 10-minute presentation.

2. Recommended Project Workflow

Phase 1 – Planning (1 day)

- Meet to agree on **business scope** (e.g., luxury cruise bookings, customers, ships, itineraries, packages, cabins, staff, payments).
- List **key entities** (Customer, Cruise, Cabin, Booking, Payment, Employee, Port, etc.).
- Decide on **team roles** above.

Phase 2 – Database Design (2 days)

- Create an **ER diagram** showing relationships.
- Identify **primary keys / foreign keys** and **functional dependencies**.
- Normalize to **3NF** (no repeating groups, remove partial & transitive dependencies).
- Example entities:
    - CUSTOMER(CustomerID, Name, Email, Phone)
    - CRUISE(CruiseID, ShipName, Destination, DepartureDate, ReturnDate, Price)
    - BOOKING(BookingID, CustomerID, CruiseID, CabinType, NumGuests, BookingDate, PaymentID)
    - PAYMENT(PaymentID, PaymentDate, Amount, Method)

Phase 3 – Implementation in SQL Server (2 days)

- Member 2 builds tables using CREATE TABLE scripts.
- Populate with realistic sample data (≈10–15 rows per table).
- Test joins & relationships.

Phase 4 – Develop Queries (2–3 days)

You need **≥10 queries** total:

- **At least 4 multi-table joins** (e.g., show bookings with customer & cruise info).
- **At least 4 aggregations/subqueries** (e.g., average revenue per cruise, total guests per destination).
- Examples:
    1. Total number of passengers booked per cruise.
    2. Highest-revenue destination.
    3. Customers with multiple bookings.
    4. Average cabin price per ship.
    5. Cruises departing in a given month.
    6. Subquery: customers spending above average.
    7. Staff who handled more than 5 bookings.
    8. Payment methods share by %.
    9. Bookings made last 30 days.
    10. Cruise with maximum occupancy.

Each query should have a **managerial interpretation** ("This helps management identify top-selling cruises").

Phase 5 – Presentation Slides (1 day)

Structure:

1. **Title & Team Intro**
2. **Project Objective & Scenario**
3. **ER Diagram / Data Model**
4. **Assumptions & Functional Dependencies**
5. **Sample Tables (screenshots)**
6. **10 SQL Queries + Results (snippets)**
7. **Managerial Insights (what each query shows)**
8. **Favorite Part / Reflection**
9. **Conclusion**

💡 Tip: use screenshots from Visual Studio's "Results" pane for query output.

Phase 6 – Rehearse (1 day)

- 10 minutes total:
    - **3–4 min:** overview + data model (Member 1)

- **4–6 min:** SQL queries + results + insights (Member 2)
- **1 min:** favorite part + wrap-up (Member 3)

🧮 3. Deliverables Checklist

Deliverable	Description	Responsible
ER Diagram (3NF)	PDF/image for slides	Member 1
SQL Code File	CREATE, INSERT, SELECT scripts	Member 2
Sample Data	For realistic query output	Member 2
10 Queries + Screenshots	≥4 joins + ≥4 aggregations	Member 2
Presentation Slides (PPT)	8–9 slides ≈ 10 min	Member 3
Submission	Upload 1 copy to Brightspace by Nov 5	Any member

🎯 Evaluation Alignment (from rubric)

Project 1

Criteria	Weight	What to Emphasize
Managerial Significance (40%)	Show real-world insights from queries.	
Database Skills (40%)	Correct data model + functional SQL.	
Presentation (20%)	Team coordination, professional slides, clear examples.	

What "Managerial Interpretation" Means

When you show an SQL query, don't just say *"this lists customers who booked a cruise"* —
add a **business insight**:

"This helps management identify repeat customers and tailor loyalty promotions."

Your SQL queries are **tools for decision-making**, not just database lookups.
So, after every query, you'll briefly answer:

🧠 "What business or operational insight does this query give to managers?"

⚓ Example Queries + Managerial Insights

(You can adjust names or fields to match your database model)

#	Example SQL Query	Managerial Interpretation (Insight)
1	**List all bookings per cruise with total revenue.**	Helps management identify **which cruise packages generate the highest income**, guiding future marketing and pricing strategies.
2	**Find top destinations by total number of passengers.**	Shows **most popular travel routes**, useful for **seasonal scheduling and fleet allocation**.
3	**Calculate average cabin price per cruise ship.**	Allows managers to **compare profitability** across ships and ensure consistent pricing tiers.
4	**List customers who made more than one booking in a year.**	Identifies **loyal or repeat customers** for targeted loyalty programs or early-bird offers.
5	**Find total payments received by month.**	Supports **financial forecasting and cash-flow analysis** for budgeting and resource planning.
6	**Subquery: List customers spending above the average fare.**	Helps marketing design **premium membership offers** for high-value clients.
7	**Show occupancy rate per cruise (booked cabins ÷ total cabins).**	Reveals **capacity utilization** — key for optimizing cruise schedules and promotions for under-booked trips.
8	**Aggregate payment methods (Credit vs. Debit vs. Online).**	Identifies **preferred payment channels**, guiding partnerships (e.g., with credit card providers).
9	**List cruises departing within the next 30 days.**	Aids operations in **crew scheduling, provisioning, and customer check-in planning.**
10	**Find the staff who processed the most bookings.**	Provides a **performance metric** for employee evaluation and potential incentives.

🧩 **How to Present This on Slides**

Slide Element	What to Include
Query Title	e.g., "Top Revenue-Generating Cruises"
SQL Snippet	A short readable version (2–3 lines)
Output Screenshot	A snippet from SQL Results pane
Managerial Insight (1 sentence)	"This query helps management identify the most profitable cruise packages to prioritize in marketing."

💡 **Tip:** Keep insights **action-oriented** — use verbs like *monitor, identify, evaluate, optimize, improve, allocate,* etc.

Example Slide Format (for one query)

Slide Title: Cruise Revenue Analysis

SQL Used:

```
SELECT CruiseName, SUM(PaymentAmount) AS TotalRevenue

FROM Booking JOIN Cruise USING (CruiseID)

GROUP BY CruiseName

ORDER BY TotalRevenue DESC;
```

Result (snippet):

CruiseName	TotalRevenue
Caribbean Paradise	$1,250,000
Mediterranean Dream	$980,000

Managerial Interpretation:
"This helps management pinpoint high-performing cruises for marketing investment and strategic pricing."